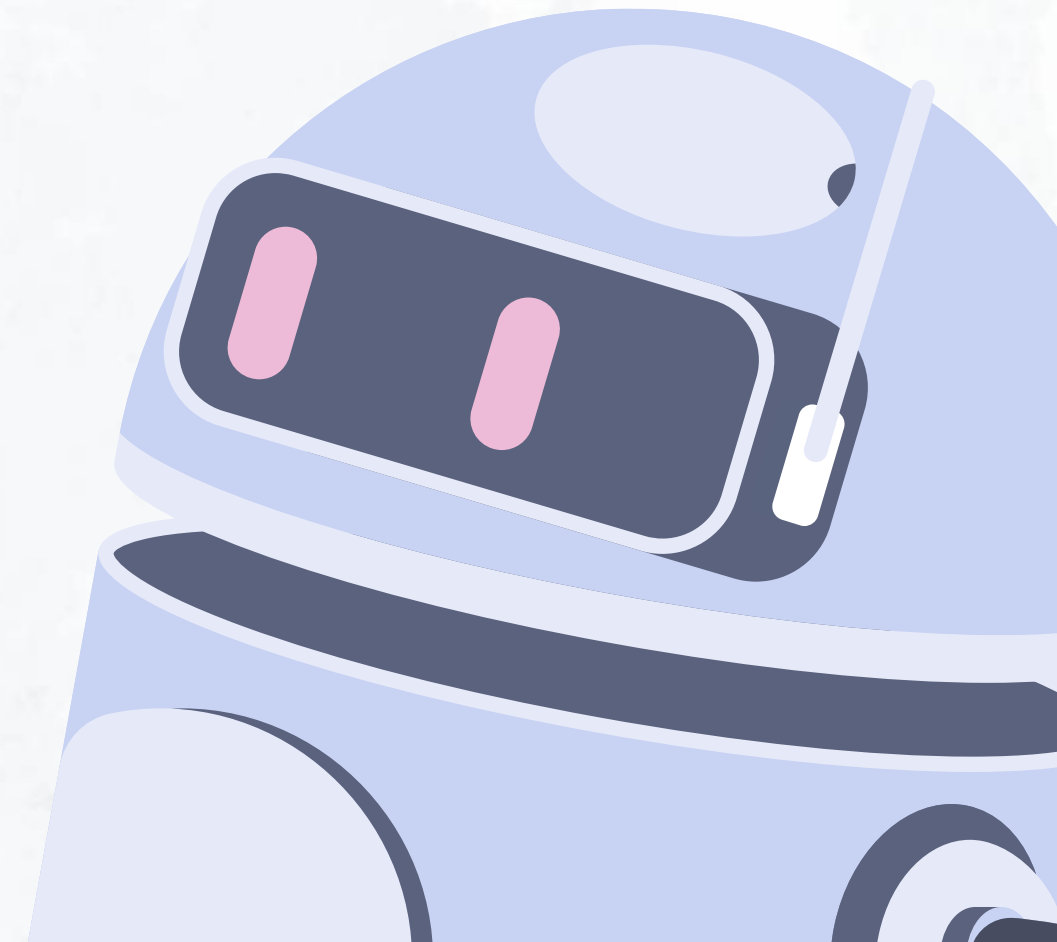
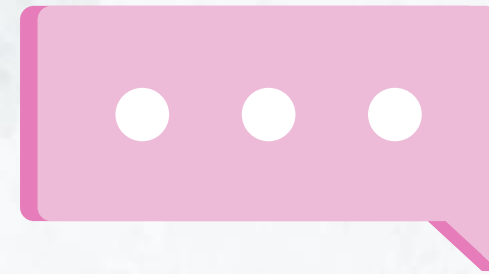


Presentation phase-3

Big Data Programming: Python

HushHush Recruiter

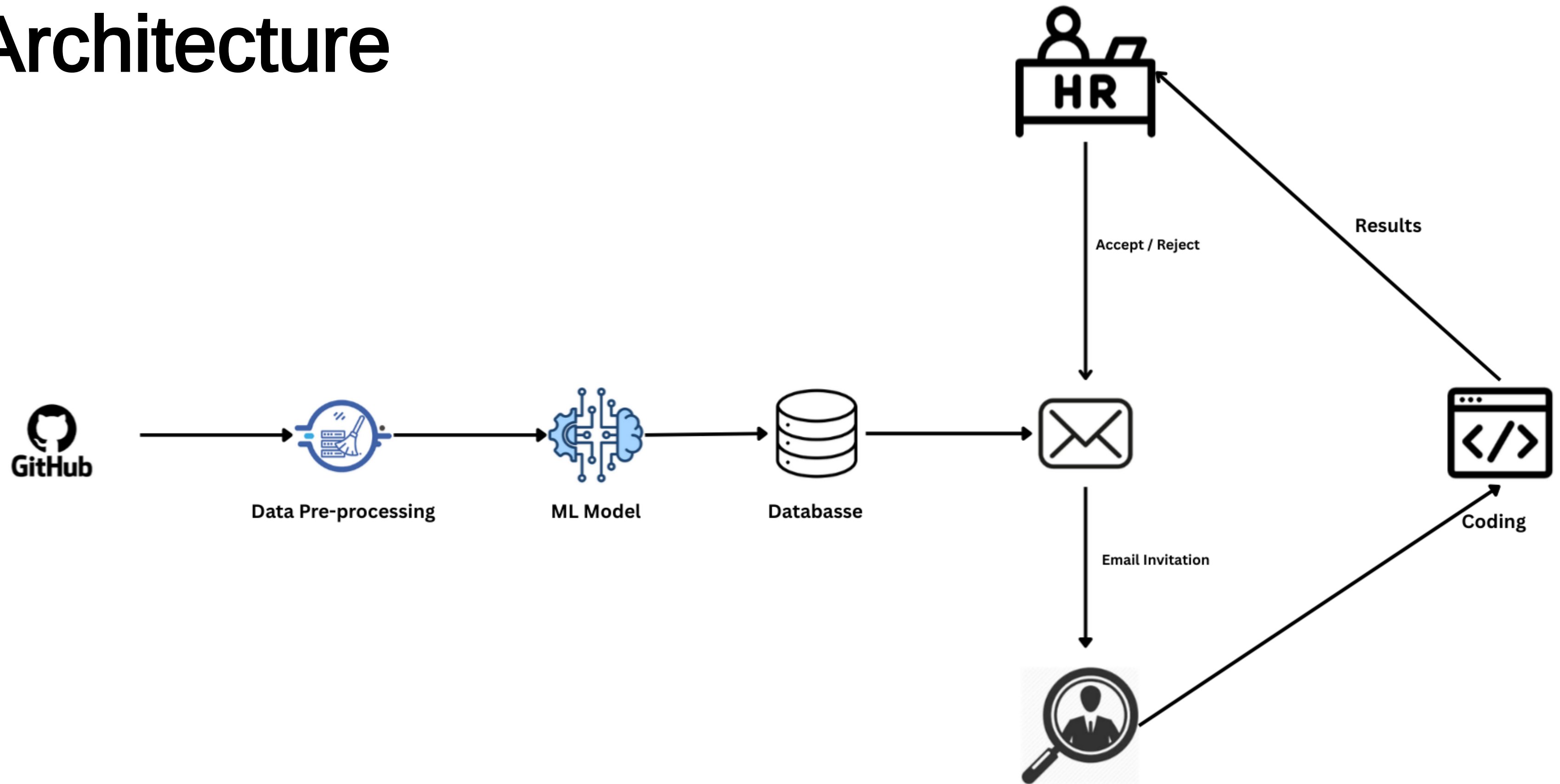
Submitted by Group-10



Team Members

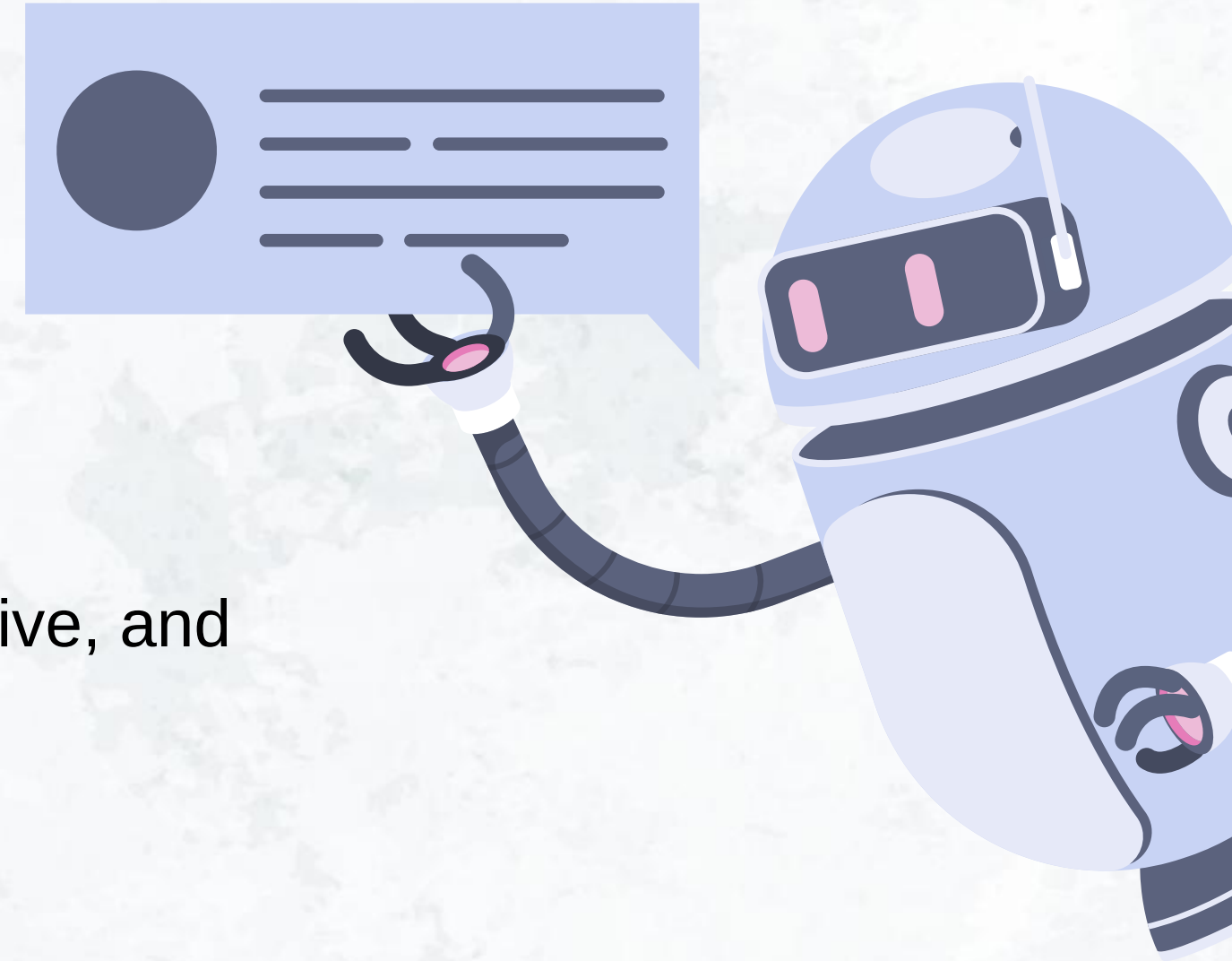
- Amith Gowda Eranna
- Darshan Gowda Budanur Madappa
- Gowrika Mahesh
- Madhumitha Ramakrishnan
- Prajwal Basavaraj

Architecture



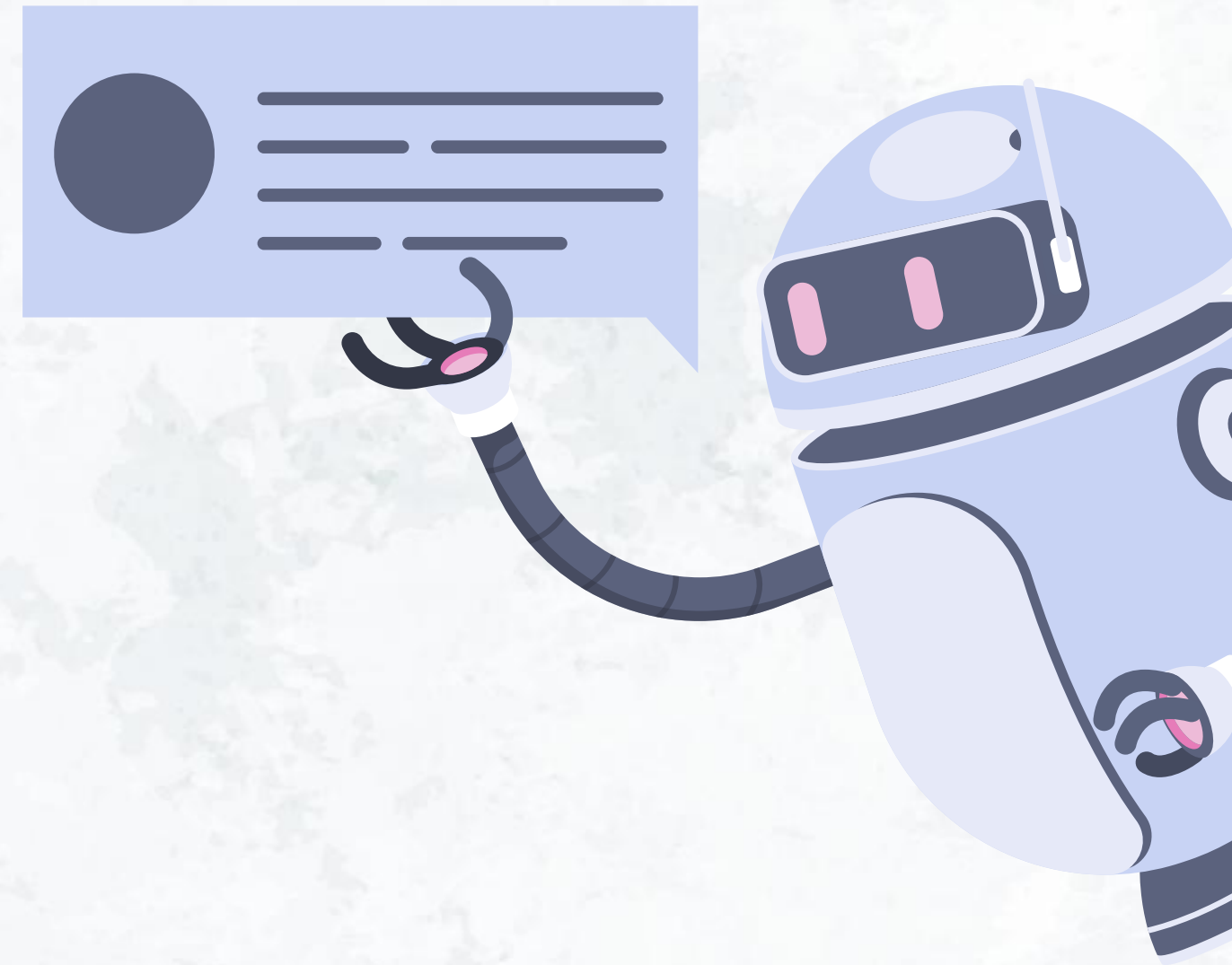
Data Source

- Why Github?
- Recruiters can see commits, issues, pull requests, and how candidates contribute to real projects.
- Provides evidence of proficiency in programming languages, frameworks, and tools.
- Active participation in open-source projects signals passion, initiative, and community engagement.
- It can also indicate how well a candidate can adapt to large, distributed projects.

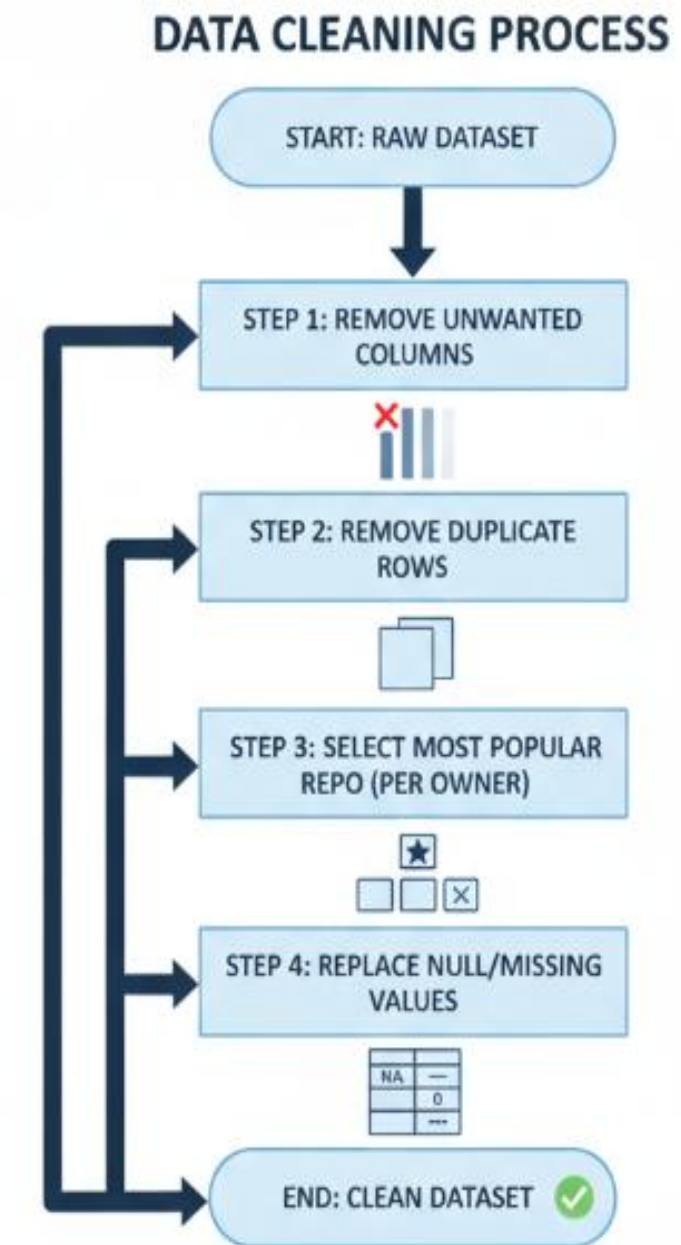
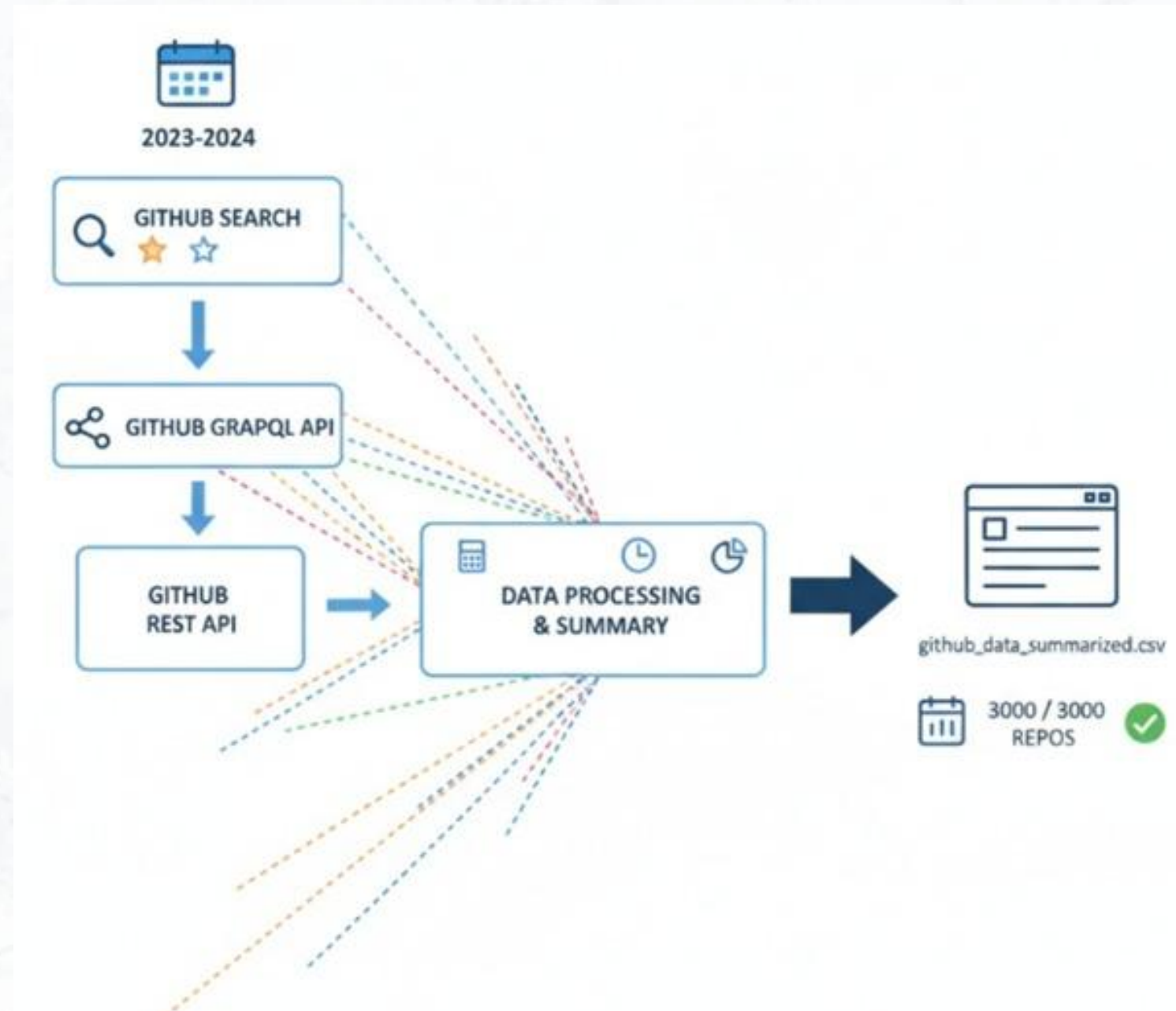


Data Extraction

- We used REST API to filter repositories by stars and fetch the latest commit author's email, acting like a wide-net search.
- GraphQL was used to retrieve all relevant data in a single query with only requested fields, handling nested relationships efficiently.



Data Pre-processing



Flow Diagram of data pre-processing

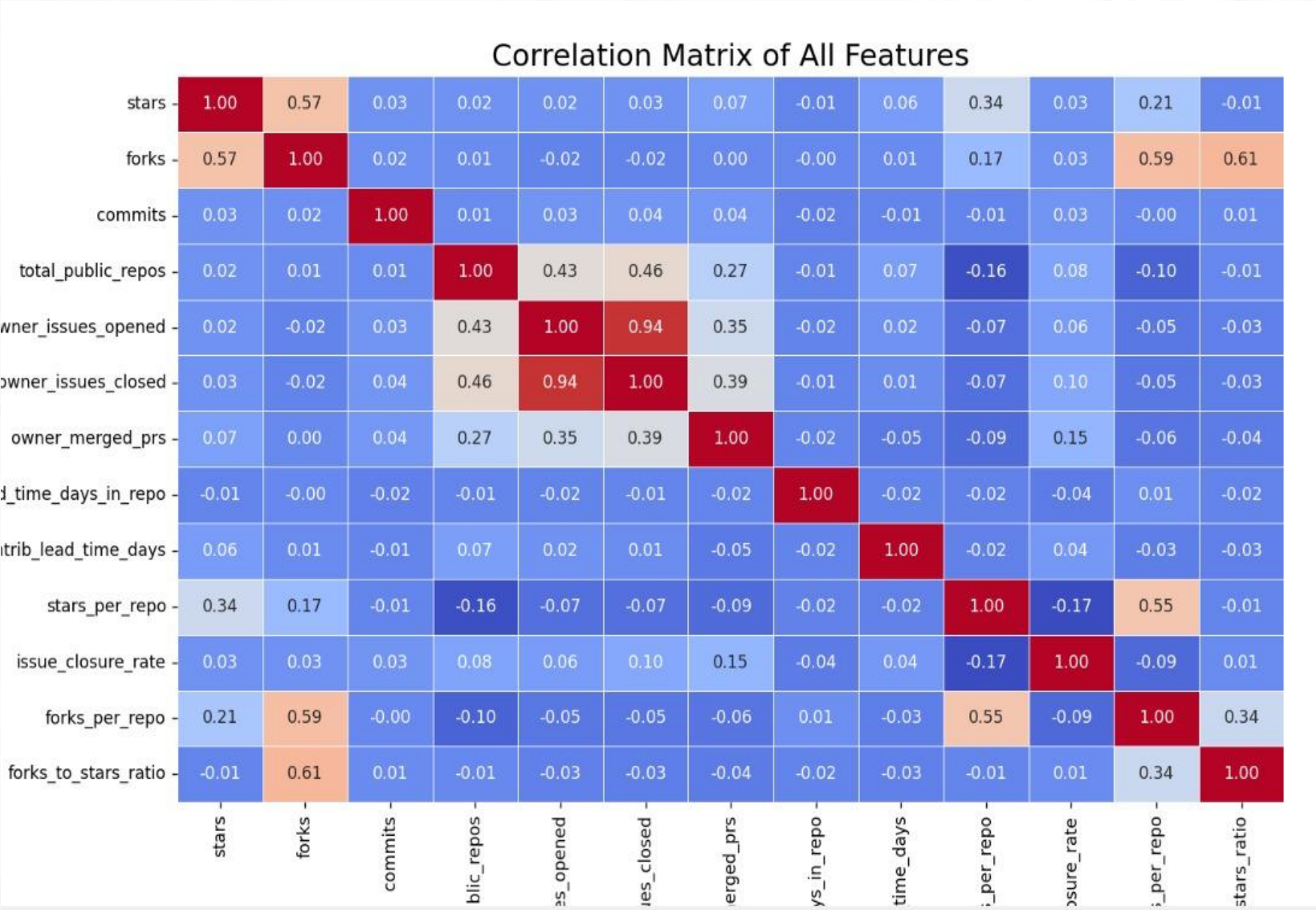
Data Pre-Processing

Feature Engineering

- **Feature 1: Responsibility Score (issue_closure_rate)**
Measures how often a developer closes the issues they open higher score = more responsible and effective problem solver.
- **Feature 2: Developer's Developer Score (forks_to_stars_ratio)**
Shows if a project is not just popular (stars) but actively built upon (forks) higher ratio = more practical and useful tool for developers

Normalization

- We scaled all numerical features to a 0–1 range using Min-Max normalization.
- Identifier columns like owner and email were kept unchanged.

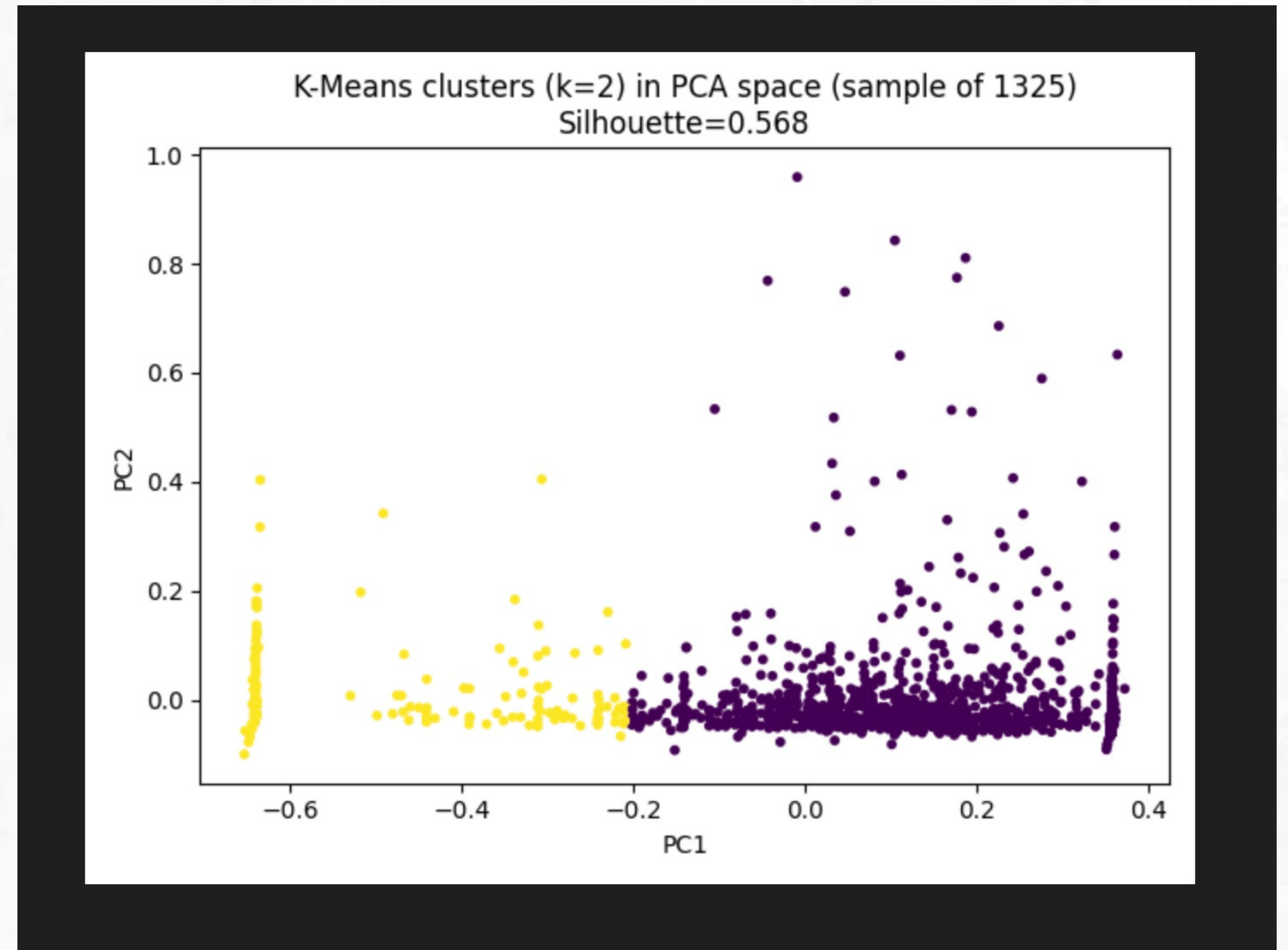


Heat Map

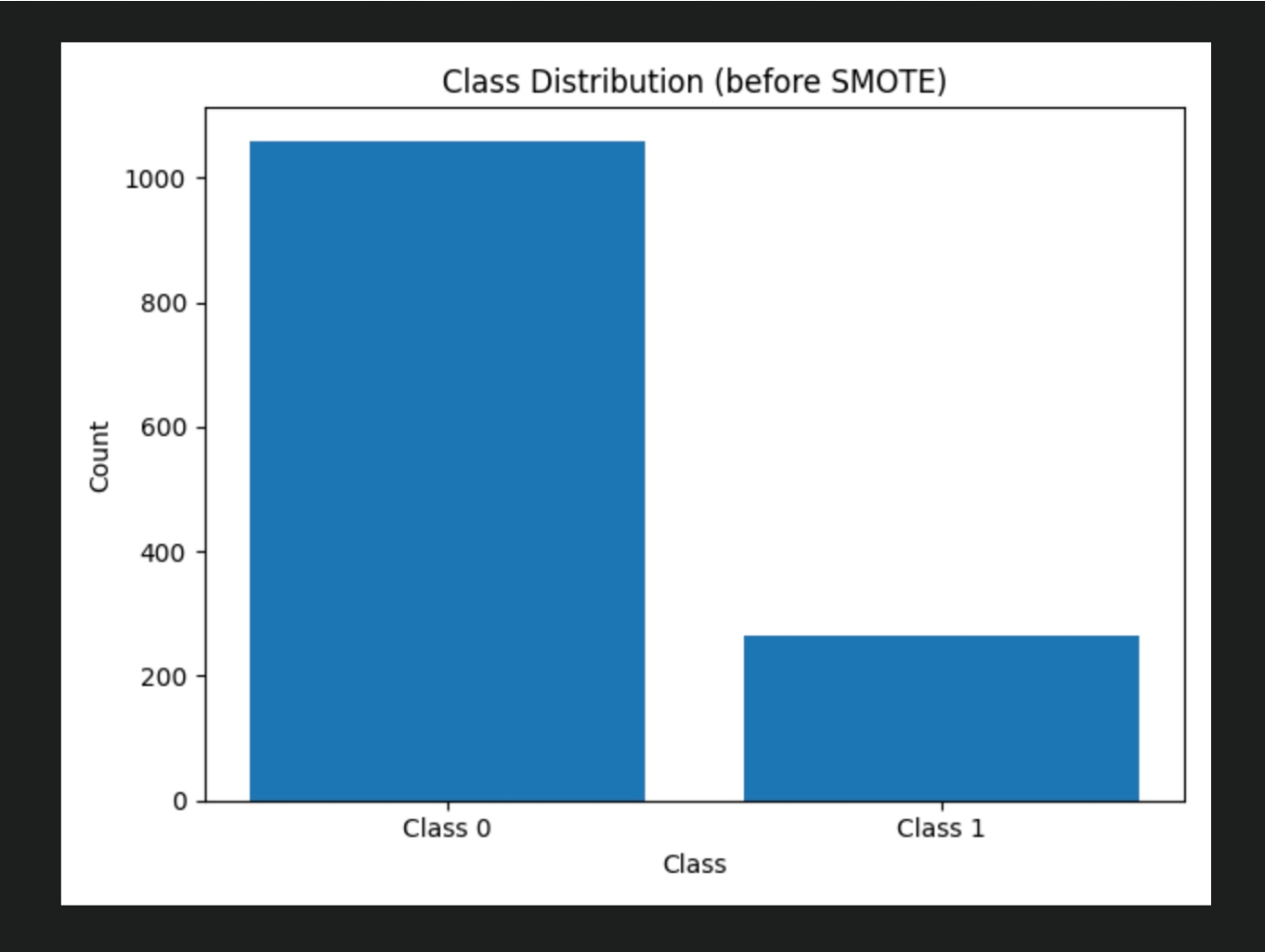
ML and Statistics

Clustering

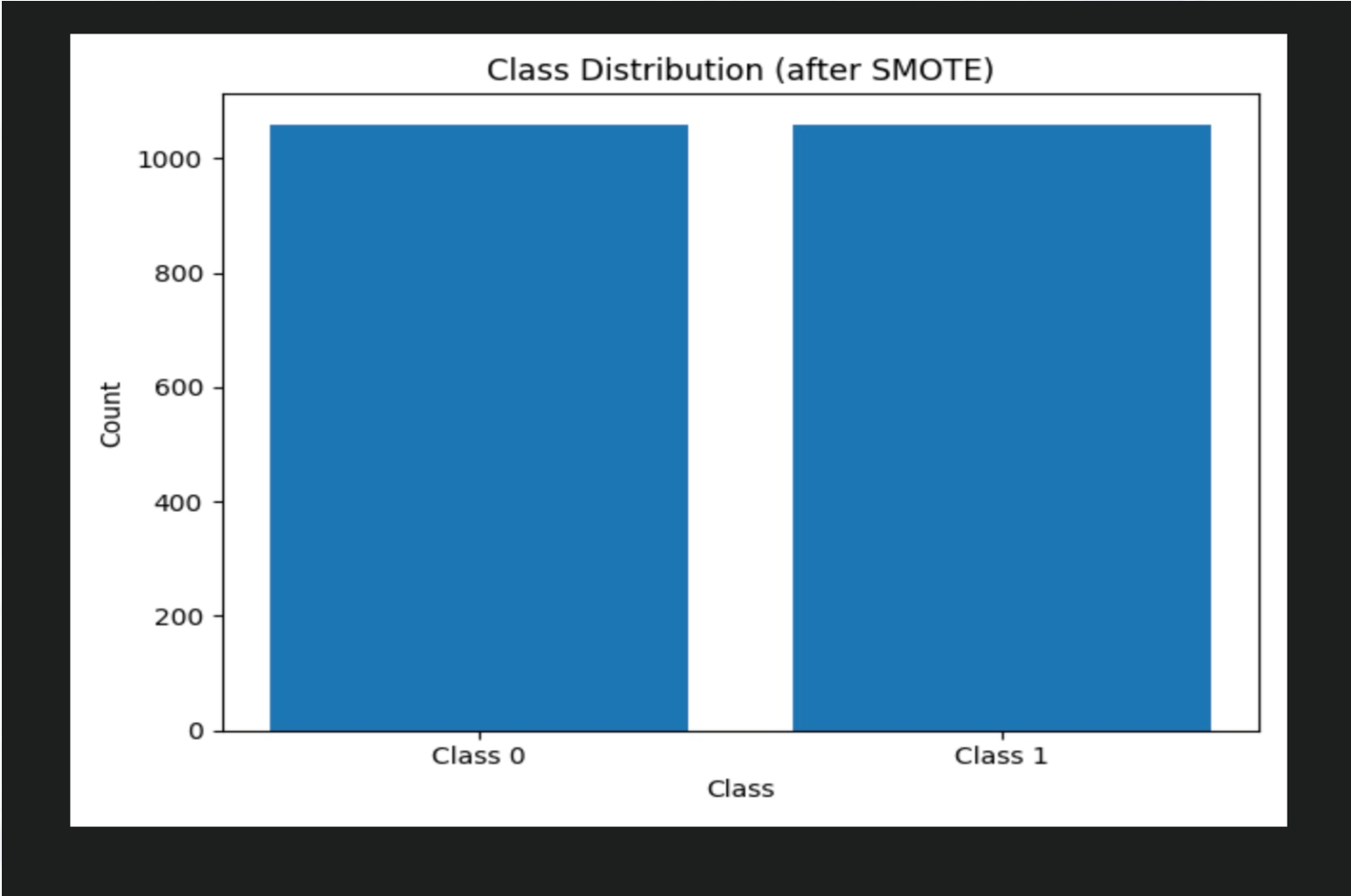
- We applied **K-Means clustering** to categorize candidates into two groups: *good* and *bad*.
- Since the dataset was imbalanced, we used the **SMOTE technique** for oversampling.



Smote



Before Smote



After Smote

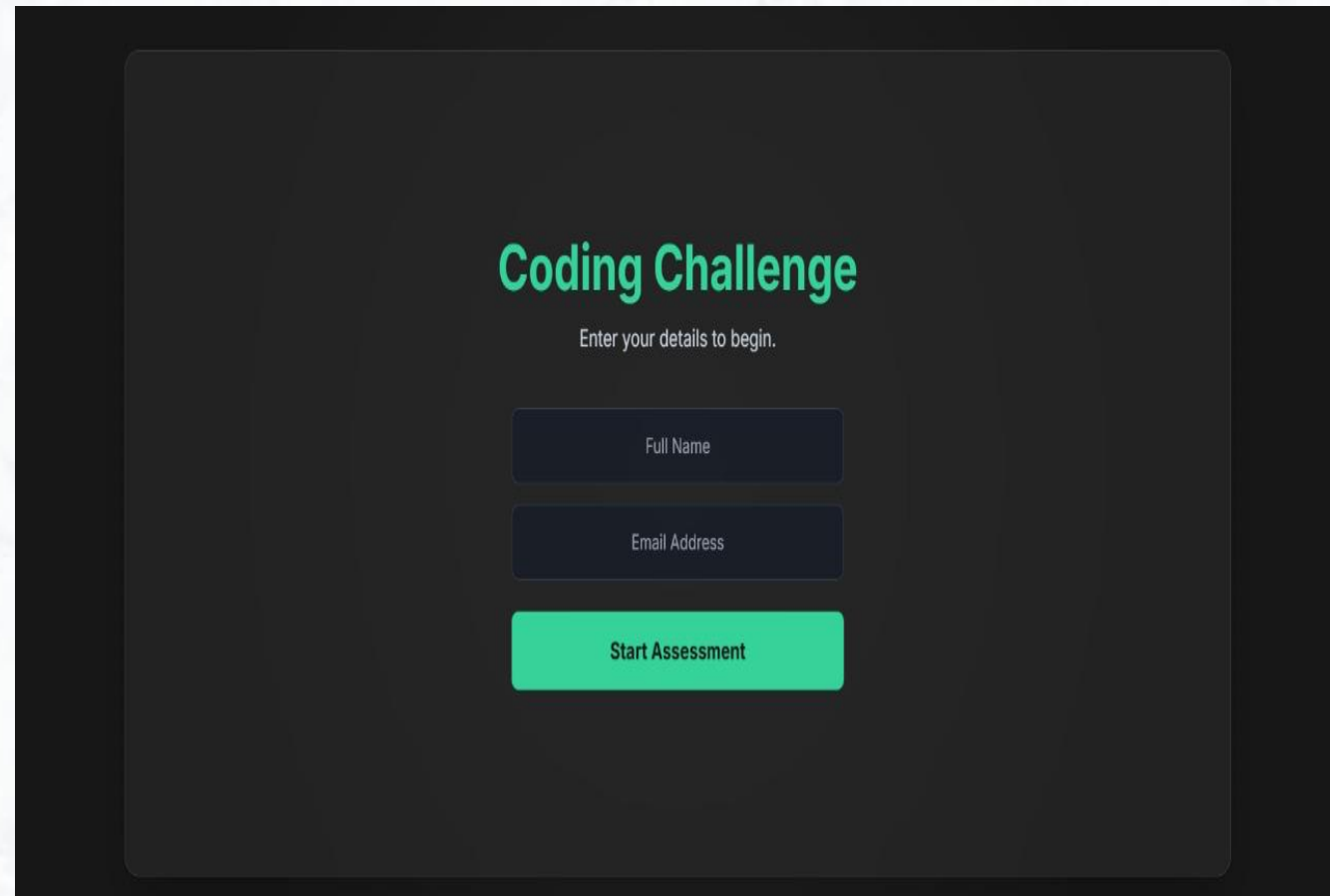
Supervised Model for Candidate Selection

- Goal: Identify top candidates from dataset based on predicted success probability.
- Model Used: **Logistic Regression with cross-validation (StratifiedKFold)**.
- Logistic Regression was ideal because it provides probability scores for each candidate. These probabilities allowed us to rank candidates and select the top 25 effectively.

```
ROC AUC: 0.8609
Average Precision: 0.8455

Accuracy (threshold=0.4132): 0.7972
Confusion matrix [tn, fp; fn, tp]:
[[171  41]
 [ 45 167]]
```

Coding Environment



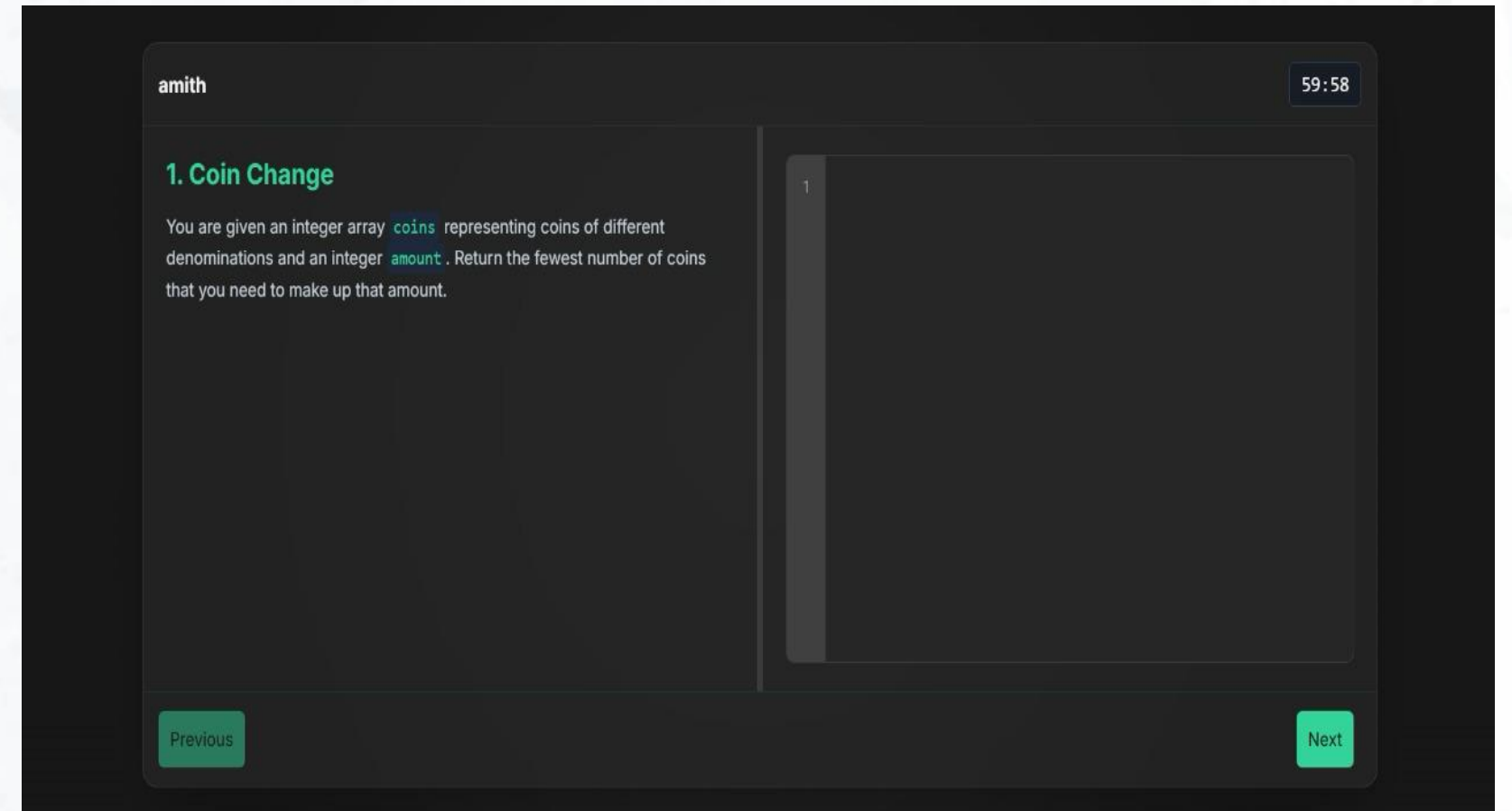
A dark-themed registration form titled "Coding Challenge" in green. Below the title, it says "Enter your details to begin." There are three input fields: "Full Name", "Email Address", and a green "Start Assessment" button.

Coding Challenge
Enter your details to begin.

Full Name

Email Address

Start Assessment



A dark-themed problem view for the "Coin Change" challenge. At the top, it shows the username "amith" and a timer "59:58". The problem description is in green text. Below the description is a code editor with a line number "1". At the bottom, there are "Previous" and "Next" buttons.

amith 59:58

1. Coin Change

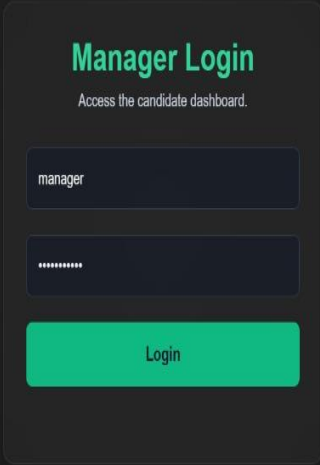
You are given an integer array `coins` representing coins of different denominations and an integer `amount` . Return the fewest number of coins that you need to make up that amount.

1

Previous Next

- UI where participants enter details and launch the challenge.
- Problem view with timer and code editor, plus simple navigation for moving to next task.

HR-Console



The Manager Login form is centered on a dark background. It features a title 'Manager Login' in green, followed by the instruction 'Access the candidate dashboard.' Below this are two input fields: the first contains the text 'manager' and the second is filled with dots. A green 'Login' button is positioned at the bottom of the form.

Candidate Submissions				Refresh
Candidate Name	Email	Score	Action	
amith	100004281@stud.srh-university.de	0	Accept	Reject
bhargav	100004281@stud.srh-university.de	275	Accept	Reject
amith	100004281@stud.srh-university.de	100	Accept	Reject
niithin	amithgowda2002@gmail.com	17	Accept	Reject
bhargav	amithgowda2002@gmail.com	0	Accept	Reject
komal	komalgd3@gmail.com	17	Accept	Reject
bhargav	cheatexams5@gmail.com	0	Accept	Reject
gowrikamahesh	gowrikamahesh2017@gmail.com	50	Accept	Reject
amith	100004281@stud.srh-university.de	0	Accept	Reject

- The HR Console provides a manager login and a centralized dashboard of candidate submissions with name, email, score, and status.
- HR can accept or reject the candidates based on scores and the candidates were notified by email.

Future Focused Enhancements

- Automate Data Fetch process.
- Fine Tune Machine Learning Algorithm Accuracy by feeding selected / rejected candidates after evaluation.
- Implement Threading for data fetching process.
- Deploy the project online.
- Use docker sandbox environment for coding platform



Thank You

